

A case study of scientific computing with limited precision numbers: Applying rescaling and baseline shifting techniques to a spectral Vlasov–Maxwell code

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ABSTRACT

We have developed *SpectraVla*, a code that solves the multi-dimensional Vlasov–Maxwell equations using the spectral method described in G. L. Delzanno [J. Comp. Phys. **301**, 338 (2015)]. This code is designed to take advantage of a chip-scale tensor core optimized for 3D convolution with 8-bit fixed-point numbers. The code is benchmarked against theory on a number of fundamental plasma physics problems. We employ rescaling and baseline shifting techniques to ensure proper precision of the results when using 8-bit fixed-point numbers.

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I. INTRODUCTION

Modern society places an accelerating demand for high speed and high energy efficiency computation, with scientific computing and artificial intelligence (AI) as two prominent examples. This calls for hardware and software innovation beyond traditional Moore’s law for semiconductor progress, which is coming to its end. The substantial energy requirements of machine learning accentuate the need for computational efficiency. For example, in 2023, Meta’s data centers consumed nearly 15 TWh,¹ while Google’s usage exceeded 24 TWh.² Microsoft reported a similar figure, consuming close to 24 TWh (primarily from renewable energy)—enough to power Paris for 2 years.³ Consequently, improving the speed and energy efficiency of modern computation becomes essential for advancing AI sustainably.

One approach is to search for novel, efficient computing substrates that can accelerate vector operations in general. A new group of hardware under development is analog in nature, and they show promise in breaking the current bottlenecks in computation speed and energy efficiency. In particular, photonics have long been thought of as a promising technology for such a substrate. Indeed, photonic devices offer large bandwidth, low power consumption, and high-speed operations.^{4–7} With multiple degrees of freedom, photonic computing is a

leading candidate for high-capacity, low-latency, high-speed matrix processing in the “More than Moore” era.⁸ This has inspired a dedicated research program to develop a chip-scale hyperspectral photonic tensor cores for low-power, ultrafast vector operations including convolution.⁹

Convolution is a fundamental operation used across diverse fields such as signal processing, image processing, computer vision, physics, and numerical simulations in general. It also plays a central role in many machine learning architectures, driving advancements in applications such as medical imaging, video analysis, autonomous driving, and physics-based modeling.^{10–13} As machine learning’s impact grows in these fields, the demand for efficient and energy-saving convolution calculations has become increasingly critical. However, using these analog substrates to natively perform convolution and other vector operations is far from an assured improvement. System and algorithm co-design is an important part of the innovation. Several key questions about the substrate need to be carefully considered for scientific numerical simulations:

- Precision: These devices are likely to provide only 8–12 bits of precision, and more bits may reduce operation speed. As a result,

rounding errors accumulate faster than using, say, double-precision operands.

- Range: In addition, the operands use fixed-point representation to cover necessary range of simulation requires additional coding effort.
- Other bottlenecks: Finally, depending on the arithmetic intensity of the code, the overall computing system may be bottlenecked by memory access and unable to operate at theoretical peak throughput.

As a concrete example for careful analysis of the computing performance of this new hardware, we have developed a spectral Vlasov–Maxwell code *SpectaVla* using the spectral method developed by Delzanno.¹⁴ Vlasov–Maxwell codes¹⁵ are a general kinetic plasma physics modeling tool that has wide applications in plasma physics. The spectral method employs convolutions in the Fourier space, which dominate the computation load.¹⁴ The fast Fourier transform (FFT) is a widely used method for accelerating convolution operations. For the convolution of two three-dimensional arrays of size $N_T = N_x \times N_y \times N_z$, the time complexity of direct convolution is $O(N_T^2)$. In contrast, a single 3D FFT of the same size has the time complexity of $O(N_T \log N_T)$. A convolution using FFT requires two forward FFTs and one inverse FFT. Considering the overhead of additional data transfer, using FFT to compute convolution offers increasingly significant advantages for large problem sizes.

In this paper, we will first show that *SpectaVla* provides correct physics results through benchmarks with theory on a wide range of plasma physics problems. We will also show that *SpectaVla* devotes most of its computation cycles to convolutions and thus can potentially benefit from the photonic tensor core to a high degree. To overcome the limitation associated with low operand precision and range on this kind of hardware, we have developed rescaling and baseline shifting techniques, which dynamically keep numerical representation of physical quantities within a range that minimizes impact of the hardware limitation. We will show that when these techniques are applied judiciously, the same overall simulation accuracy as traditional double-precision simulations can be achieved with 8-bit fixed-point precision and virtually no overhead over a wide range of physics problems.

This work is organized as follows: Sec. II provides a brief introduction to the code. The implementation of limited-precision convolution is detailed in Sec. III. Section IV presents benchmarking of the code against theory. The time breakdown of the code is discussed in Sec. V. Finally, Sec. VI concludes with a summary and discussion.

II. EQUATIONS, SCALING, AND SOLVER IMPLEMENTATION

SpectaVla solves the dimensionless Vlasov–Maxwell equations for multiple species

$$\begin{aligned} \frac{\partial f_s}{\partial t} + \mathbf{v} \cdot \nabla_{\mathbf{x}} f_s + q_s \frac{\omega_{cs}}{\omega_{pe}} (\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \nabla_{\mathbf{v}} f_s &= 0, \\ \frac{\partial \mathbf{B}}{\partial t} &= -\nabla \times \mathbf{E}, \\ \frac{\partial \mathbf{E}}{\partial t} &= \nabla \times \mathbf{B} - \frac{\omega_{pe}}{\omega_{ce}} \sum_s q_s \int_{-\infty}^{+\infty} \mathbf{v} f_s d\mathbf{v}, \\ \nabla \cdot \mathbf{E} &= \frac{\omega_{pe}}{\omega_{ce}} \sum_s q_s \int_{-\infty}^{+\infty} f_s d\mathbf{v}, \\ \nabla \cdot \mathbf{B} &= 0. \end{aligned}$$

In this set of equation, f_s represents the distribution function for species s with the normalization of $\int f_s d\mathbf{v} = n_s$, which is the species density normalized to a reference electron density n_0 ($s = e, i$ representing electrons and ions, respectively). Time t is normalized by the electron plasma frequency ω_{pe} at n_0 , velocity $\mathbf{v}$ by the speed of light c , and length by c/ω_{pe} . $\mathbf{E}$ and $\mathbf{B}$ are the normalized electric and magnetic fields, with $\mathbf{B}$ normalized by a reference magnetic field B_0 and $\mathbf{E}$ by cB_0 , respectively. The particle charge q_s is normalized by the electron charge e , and $\omega_{cs} \equiv eB_0/m_s$, with m_s being the particle mass for species s .¹⁴

By employing the fully implicit spectral method based on a truncated Hermite expansion in velocity space and a Fourier representation in space, combined with Crank–Nicolson time discretization described in Ref. 14, the multi-dimensional Vlasov–Maxwell equations can be reduced to a set of nonlinear equations in the residual form,

$$R(X) = 0, \quad (1)$$

where $X = [C_s; B_x; B_y; B_z; E_x; E_y; E_z]$, and the subscripts x, y , and z for B and E denote the three spatial dimensions. C_s represents the coefficients obtained by expanding the distribution function f_s using Hermite polynomials in the Fourier space and is a six-dimensional array in the Fourier–Hermite space. Each element of C_s is represented as $C_{s,n,m,p}^{k_x,k_y,k_z}$, where (n, m, p) are indices in the Hermite space, and (k_x, k_y, k_z) are indices in the Fourier space. Specially, $C_{s,n,m,p}$ is a three-dimensional array consists of elements $C_{s,n,m,p}^{k_x,k_y,k_z}$. Similarly, the field components, such as E_x , are three-dimensional arrays, with individual elements denoted as $E_x^{k_x,k_y,k_z}$. The detailed construction and derivation of $R(X)$ can be found in Ref. 14. We use bold symbols to represent physical quantities in the ordinary space. For instance, $\mathbf{E}_x$ denotes the electric field along the x -direction in the ordinary space, and $E_x \equiv \text{FFT}[\mathbf{E}_x]$, where FFT stands for Fast Fourier Transform.

We developed *SpectaVla* to solve the residual equation [Eq. (1)], using the inexact Jacobian–Free Newton–Krylov (JFNK) method with preconditioned Generalized Minimum Residual (GMRES) algorithm, following the approach in Delzanno’s work.¹⁴ The GMRES solver in our code is adapted from MATLAB’s gmres function.¹⁶

III. CONVOLUTION SOLVERS

The computation load in solving Eq. (1) is dominated by 3D convolutions. In this code, two convolution solvers are implemented. The first solver uses FFT to calculate the circular convolution of A and D , denoted as $A * D$ in Eq. (2),

$$A * D = \text{IFFT}[\text{FFT}[A] \times \text{FFT}[D]], \quad (2)$$

where IFFT stands for inverse FFT. In this paper, we refer to this solver as the FFT solver. The results from this solver serve as a benchmark for the second solver, which computes the 3D circular convolution directly using its definition

$$[A * D]^{i_x, i_y, i_z} = \sum_{i'_x=0}^{N_x-1} \sum_{i'_y=0}^{N_y-1} \sum_{i'_z=0}^{N_z-1} A^{i_x-i'_x, i_y-i'_y, i_z-i'_z} D^{i'_x, i'_y, i'_z}. \quad (3)$$

Here, the superscript (i_x, i_y, i_z) and (i'_x, i'_y, i'_z) correspond to the spatial indices, and N_x, N_y , and N_z represent the number of elements in each spatial dimension. Negative indices of A are adjusted using modular arithmetic. Specifically, when $i_x - i'_x$ is negative, it is replaced with

$i_x - i'_x + N_x$. Similarly, $i_y - i'_y$ and $i_z - i'_z$ are adjusted to $i_y - i'_y + N_y$ and $i_z - i'_z + N_z$, respectively, whenever they are negative. In this paper, we refer to the solver based on Eq. (3) as the *hardware solver*. The hardware solver's performance is expected to be significantly enhanced on the new hardware currently under development. Both Eqs. (2) and (3) are equivalent to the FFT convolution solver by Delzanno that conserves mass, momentum and energy.¹⁴ Furthermore, these two solvers are equivalent at a discrete level.

In the hardware solver, numbers are represented in an 8-bit fixed-point format, to simulate on a CPU the effects of limited precision anticipated of the new hardware on convolution. An 8-bit fixed-point format in this paper consists of, from left to right, one sign bit, one integer bit, and six bits representing the fractional part. Outside the hardware solver, double-precision numbers are used. Therefore, additional conversion is required before passing arrays to the hardware solver. To convert double-precision numbers into the numbers expected by the hardware solver, a number $\mathbb{C}$ is first multiplied by 2^6 , shifting the decimal point to the end of the 8th bit. Then, the scaled number is rounded to the nearest integer and scaled back by dividing it by 2^6 . Underflow is inherently considered during the rounding process, while overflow is checked before outputting. By employing the representation defined in this paper, the maximum value that can be expressed using an 8-bit fixed-point number is $2 - 2^{-6} = 1.9844$. Any double precision number $\mathbb{C}$ with an absolute value greater than 1.9844 will be replaced with $\text{sign}(\mathbb{C}) * 1.9844$, where sign is the sign function.

Each element in the arrays A and D passed to the hardware solver must undergo this conversion. In the Fourier space, these elements are complex numbers, and the real and imaginary parts are converted separately. Following this conversion and the convolution process using Eq. (3), the results are converted again to ensure they comply with the hardware specifications.

To prevent significant loss of precision during the conversion process, at the beginning of the simulation double-precision variables are scaled to a range that can be represented by the 8-bit fixed-point format: C_s is normalized using $C_{s,\text{scale}}$ to obtain $\tilde{C}_s$, where $C_{s,\text{scale}} = \max\{|C_{s,n,m,p}^{k_x,k_y,k_z}|\}$ at $t = 0$. Similarly, the field components, such as E_x , are scaled using scaling factors such as $2E_{x,\text{scale}}$ to get $\tilde{E}_x$, where $E_{x,\text{scale}} = \max\{|E_x^{k_x,k_y,k_z}|\}$ at $t = 0$. Scaling the field components by twice its maximum value mitigates the risk of overflow during convolution. The scaled variables ($\tilde{C}_s$, $\tilde{E}_x$, ...) are then solved from Eq. (1) and also serve as inputs to the hardware solver.

IV. SIMULATION BENCHMARK

To validate the code, five test cases were performed and the results were benchmarked: Landau damping (LD), dispersion relations of waves in a magnetized plasma (WMP) and ion acoustic waves (IAW) in a hot unmagnetized plasma, and two-stream instabilities (TSI) in the linear and nonlinear stage. The LD and IAW cases assessed the solvers' accuracy over different time scales, while the WMP cases evaluated the code's performance in scenarios where the electromagnetic fields have amplitudes of varying orders of magnitude. The TSI case in the linear stage investigated the code's performance in accurately predicting the temporal growth of instabilities. Finally, the TSI case in the nonlinear stage studied the code's capability in handling nonlinear problems. In these simulations, all physical

quantities are initialized in the ordinary space and subsequently transformed into the Fourier space. Unless stated otherwise, any unspecified quantities in Secs. IV A–IV F are initialized to zero. The tolerance for the GMRES solver was set to 10^{-6} . To evaluate the impact of limited-precision convolution, the simulations were performed using both solvers [Eqs. (2) and (3)]. The scaling procedure described in Sec. II was applied at the conclusion of the initialization phase when using the hardware solver.

A. Landau damping

The physical and numerical parameters for simulation setup are defined as follows. L represents the size of the simulation domain, n_v and n_x denote the number of modes used in the Hermite space and the Fourier space, respectively, $\alpha_s \equiv \sqrt{2}v_s$, where v_s is the thermal velocity of species s , and u_s is the drift velocity of species s . For the Landau damping problem, the following parameters were used: $L = \{2\pi, 2\pi, 2\pi\}$, $n_x = \{16, 1, 1\}$, $n_v = \{10, 10, 10\}$, $\alpha_e = \{\sqrt{2}, \sqrt{2}, \sqrt{2}\}$, $u_e = \{0.0, 0.0, 0.0\}$, $m_i/m_e = 1836.0$, $\alpha_i = \sqrt{m_e/m_i}\alpha_e$, $u_i = \{0.0, 0.0, 0.0\}$, and $\omega_{pe}/\omega_{ce} = 1.0$.¹⁴

The 1D3V simulations ran up to $\omega_{pet} = 10.0$ with a time step of $\omega_{pe}dt = 0.02$. An artificial damping rate $\mu = 10.0\omega_{pe}$ was used. The use of artificial damping is a common practice to eliminate the well-known recurrence effect caused by the finite resolution in velocity space while preserving the conservation laws.¹⁴ The physical damping rate is not affected by this artificial damping.¹⁸

At the beginning of the simulation, a small perturbation was applied to the electron distribution function in the ordinary space by setting $C_{e,0,0,0} = (1 + \epsilon \cos(x))/(\alpha_{e,x}\alpha_{e,y}\alpha_{e,z})$, the ion density as $C_{i,0,0,0} = 1/(\alpha_{i,x}\alpha_{i,y}\alpha_{i,z})$, and the electric field component as $E_x = -\epsilon \sin(x)$, where $\epsilon = 0.01$.

Figure 1 compares the evolution of the amplitude of the dominant mode $E_x^{1,0,0}$ from different simulations. The reference solution, represented by a solid blue line, was taken from Ref. 14. The solution using the FFT solver, depicted by orange hollow triangles, closely matched the reference, indicating a high level of agreement. The green dashed line in Fig. 1 represents the theoretical Landau damping rate for the mode $E_x^{1,0,0}$, obtained by numerically solving the dispersion relation

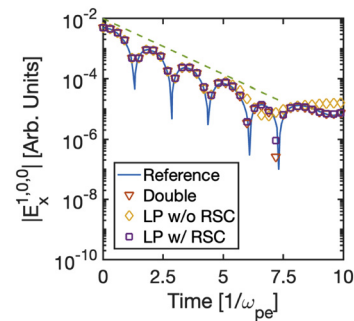


FIG. 1. Time history of the amplitude of the dominant mode $|E_x^{1,0,0}|$ in the Landau damping simulation. The green dashed line represents the theoretical Landau damping rate ($\gamma/\omega_{pe} = -0.851$). In the legend, “Double” denotes double precision, “LP” limited precision, and “RSC” rescaling.

$$1 - \frac{\omega_{pe}^2}{k^2 \alpha_{e,x}^2} Z' \left(\frac{\omega}{k \alpha_{e,x}} \right) = 0,$$

where Z denotes the plasma dispersion function.¹⁹ The frequency for the mode $E_x^{1,0,0}$ is $\omega/\omega_{pe} = 2.05 - 0.851i$, and the simulation using the FFT solver successfully reproduced the theoretical Landau damping rate ($\gamma/\omega_{pe} = -0.851$). The deviation from linear damping near $\omega_{pe}t \approx 7.5$ was due to ion motion.¹⁷

The hardware solver results, represented by yellow diamonds, began to show significant deviations after $5.0/\omega_{pe}$. The attenuation of the electric field due to Landau damping made it increasingly challenging to capture accurately with limited precision (LP), which can only represent positive numbers down to 2^{-6} . By $5.0/\omega_{pe}$, the peak $|E_x^{1,0,0}|$ value after scaling is 0.0148, which was smaller than the minimum resolution of 2^{-6} .

To enhance the performance of the hardware solver, we applied rescaling (RSC) to the original data when specific criteria were met. At the end of each time step, we calculated $C_{max} \equiv \max\{|\tilde{C}_e|, |\tilde{C}_i|\}$ and $F_{max} = \max\{|\tilde{E}|, |\tilde{B}|\}$. RSC was performed whenever C_{max} or F_{max} fell outside the acceptable range of $[0.25, 1.5]$. This RSC technique significantly improved the performance of the hardware solver (purple squares in Fig. 1) after $5/\omega_{pe}$, bringing it closer to the result using the FFT solver.

B. High-frequency waves in magnetized plasmas: Parallel modes

For 1D3V simulations of waves in plasmas with a constant external parallel magnetic field B_x , the following parameters were used: $L = \{20\pi, 20\pi, 20\pi\}$, $n_x = \{64, 1, 1\}$, $n_y = \{5, 5, 5\}$, $\alpha_e = \{0.02, 0.02, 0.02\}$, $u_e = \{0.0, 0.0, 0.0\}$, $m_i/m_e = 1836.0$, $\alpha_i = 10\alpha_e\sqrt{m_e/m_i}$, $u_i = \{0.0, 0.0, 0.0\}$, and $\omega_{pe}/\omega_{ce} = 4.0$. The background plasma was uniform at $t = 0$, and $C_{e,0,0,0} = 1/(\alpha_{e,x}\alpha_{e,y}\alpha_{e,z})$, $C_{i,0,0,0} = 1/(\alpha_{i,x}\alpha_{i,y}\alpha_{i,z})$. The background magnetic field was $B_x = 1.0$.¹⁴ The system at $t = 0$ was perturbed with uniform random noises ranging from $-\epsilon$ to ϵ in the electromagnetic fields, where $\epsilon = 10^{-5}$. The simulations ran up to $\omega_{pe}t = 500.0$ with a time step of $\omega_{pe}dt = 0.1$. Neglecting the ion contribution, the high-frequency dispersion relation for waves propagating parallel to the background magnetic field is

$$\frac{k^2 c^2}{\omega^2} = 1 - \frac{\omega_{pe}^2/\omega^2}{1 \pm \omega_{ce}/\omega}. \quad (4)$$

Figure 2(a) shows the wave spectra of $\log_{10}|E_y(k_x, \omega)|$ from the simulation using the FFT solver. In this plot, the right-hand and left-hand circularly polarized (RHCP and LHCP) waves can be clearly identified above the cutoffs at $1.13\omega_{pe}$ and $0.88\omega_{pe}$, respectively. The whistler mode can be observed below $\omega_{ce} = 0.25\omega_{pe}$. All these modes aligned well with the theoretical curves [Eq. (4)]. The LHCP resonance at ω_{ci} was not observed due to the short run time of $0.068/\omega_{ci}$. Ion dynamics will be presented in Sec. IV D.

This problem, however, cannot be simulated correctly by the hardware solver using RSC alone. The RHCP and LHCP waves cannot be differentiated and the whistler mode was not observed [Fig. 2(b)]. We define a relative error in the electric field, $|(E_y - E_{y,ref})/E_{y,ref}|$, with $|E_{y,ref}|$ being the result obtained using the FFT solver and the overline denoting the average in the Fourier space. In this simulation,

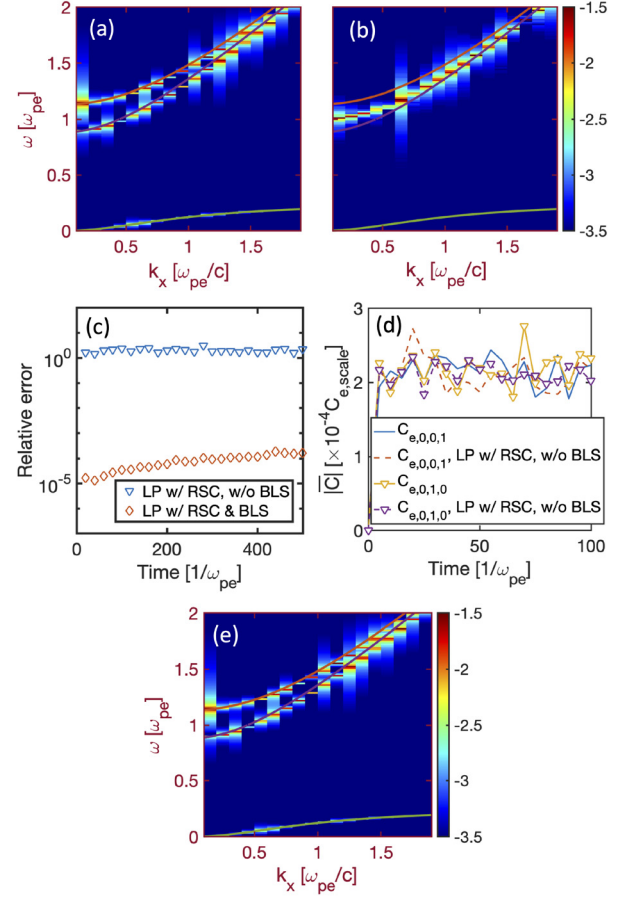


FIG. 2. Wave spectra of $\log_{10}|E_y(k_x, \omega)|$ in magnetized plasma with $B_x = 1.0$ using (a) the FFT solver, (b) the hardware solver without BLS, and (e) the hardware solver with BLS. Solid curves correspond to the dispersion relation from Eq. (4). (c) Mean relative error of $E_y(k_x, \omega)$ from the hardware solver. (d) Evolution of $|C_{e,0,0,1}|$ and $|C_{e,0,1,0}|$ in time. In the legend, “LP” represents limited precision, “RSC” denotes rescaling, and “BLS” stands for baseline shifting.

the time evolution of the relative error remained above 1 [the blue triangles in Fig. 2(c)]. This was due to disparity between large equilibrium density and its small fluctuating counterparts, which determine the dispersion relation. The dominant element in the coefficients of the distribution function was $C_{s,0,0,0}^{0,0,0}$, which scaled to 1 and remained constant in the simulation. (Constant $C_{s,0,0,0}^{0,0,0}$ reflects the conservation of particle numbers.¹⁴) Other coefficients were typically orders of magnitude smaller than $C_{s,0,0,0}^{0,0,0}$. For instance, the amplitude of $|C_{e,0,1,0}|$ remained below 3×10^{-4} [Fig. 2(d)]. When converted to the 8-bit fixed-point precision, these small values rounded to 0, preventing the system from correctly responding to the background magnetic field. For example, accurately modeling the evolution of $C_{e,0,0,1}$ requires the convolution of B_x and $C_{e,0,1,0}$. However, as $B_x * C_{e,0,1,0}$ became 0 under this precision, the solver’s ability to handle them correctly was hampered.

To improve the performance of the hardware solver in this situation, we introduced a baseline shifting (BLS) technique. In this

situation, it is natural to split the variable into a baseline and a perturbation. We expand the circular convolution $C_{s,n,m,p} * B_x$ by expressing $C_{s,n,m,p}$ and B_x as $C_{s,n,m,p} = C_{s,n,m,p}^b + dC_{s,n,m,p}$ and $B_x = B_x^b + dB_x$, where $C_{s,n,m,p}^b$ and B_x^b are baselines for $C_{s,n,m,p}$ and B_x , and $dC_{s,n,m,p}$ and dB_x are perturbations. The convolution can then be expanded as follows:

$$\begin{aligned} & (C_{s,n,m,p}^b + dC_{s,n,m,p}) * (B_x^b + dB_x) \\ &= C_{s,n,m,p}^b * B_x^b + C_{s,n,m,p}^b * dB_x + dC_{s,n,m,p} * B_x^b + dC_{s,n,m,p} * dB_x. \end{aligned}$$

In this expanded form, the baseline $C_{s,n,m,p}^b$ and B_x^b remain unchanged throughout the simulation. The first three terms are linear, as the baseline remains constant. Consequently, they do not require the convolution solver and can be computed directly outside the hardware solver. The hardware solver computes only the convolution of the two scaled perturbations. In *SpectraVla*, the BLS technique has been applied to all electromagnetic field components, even in cases where their small perturbations were identically zero. This approach enables the hardware solver to accurately address a wider range of plasma problems. Our tests confirmed that applying BLS to perturbations that were identically zero did not introduce significant computational overhead, ensuring efficient performance without compromising accuracy.

For this problem, the baselines for the electron and ion density were set to the initial background distribution functions, while the baseline for B_x was set to 1.0. After applying this technique, the mean relative error in Fig. 2(c) remained below 5×10^{-4} , and all characteristic modes were correctly captured under limited precision in Fig. 2(e).

C. High-frequency waves in magnetized plasmas: Perpendicular modes

By changing the direction of the external magnetic field from parallel to perpendicular ($B_x = 0$, $B_y = 1.0$) while keeping other settings the same, we were able to model perpendicular modes. The high-frequency extraordinary waves have the following dispersion relation:

$$\frac{k^2 c^2}{\omega^2} = 1 - \frac{1}{\omega^2} \frac{\omega_{pe}^2 (\omega^2 - \omega_{pe}^2)}{(\omega^2 - \omega_{pe}^2) - \omega_{ce}^2}. \quad (5)$$

Figure 3(a) shows the wave spectra of $\log_{10}|E_x(k_x, \omega)|$ from the simulation using the FFT solver. The extraordinary modes above the cutoff of $1.13\omega_{pe}$ and below the upper hybrid resonance $\omega_{UH} = 1.02\omega_{pe}$ were clearly identified and validated by the theoretical dispersion relation in Eq. (5). The BLS technique was also important in this problem. Without this technique, the hardware solver did not capture the response from the background density to the magnetic field [Fig. 3(b)], and the mean relative error was high in Fig. 3(c). After applying BLS, the modes were correctly modeled, and agreed with the theoretical dispersion relation [Fig. 3(d)]. The mean relative error in this case was below 10^{-3} in Fig. 3(c).

D. Ion acoustic waves in a hot unmagnetized plasma

As an example of ion dynamics, we simulated ion acoustic waves (IAW). In these 1D3V simulations, the following parameters were used: $L = \{20\pi, 20\pi, 20\pi\}$, $n_x = \{64, 1, 1\}$, $n_y = \{5, 5, 5\}$, $\alpha_e = \{\sqrt{2}, \sqrt{2}, \sqrt{2}\}$, $u_e = \{0.0, 0.0, 0.0\}$, $m_i/m_e = 1836.0$, $\alpha_i = 0.1\alpha_e\sqrt{m_e/m_i}$, $u_i = \{0.0, 0.0, 0.0\}$, and $\omega_{pe}/\omega_{ce} = 1.0$. The

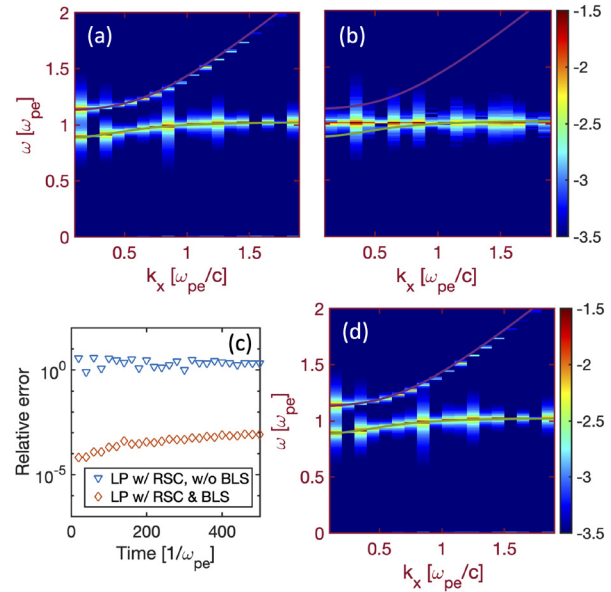


FIG. 3. Wave spectra of $\log_{10}|E_x(k_x, \omega)|$ in magnetized plasma with $B_y = 1.0$ using (a) the FFT solver, (b) the hardware solver without BLS, and (d) the hardware solver with BLS. Solid curves correspond to the dispersion relation from Eq. (5). (c) Mean relative error of $E_x(k_x, \omega)$ from the hardware solver. In the legend, “LP” represents limited precision, “RSC” denotes rescaling, and “BLS” stands for baseline shifting.

background plasma was uniform at $t = 0$, and $C_{e,0,0,0} = 1/(\alpha_{e,x}\alpha_{e,y}\alpha_{e,z})$, $C_{i,0,0,0} = 1/(\alpha_{i,x}\alpha_{i,y}\alpha_{i,z})$. The system at $t = 0$ was perturbed with uniform random noises ranging from $-\epsilon$ to ϵ in E_x , where $\epsilon = 10^{-2}$. The simulations ran up to $\omega_{pe}t = 10000.0$ with a time step of $\omega_{pe}dt = 0.1$. An artificial damping rate $\mu = \omega_{pe}$ was used.

Under the given plasma conditions, the electron Landau damping strongly damped electron plasma waves with large k . As a result, IAW dominated in these simulations. Figure 4(a) shows the wave spectra of $\log_{10}|E_x(k_x, \omega)|$ from the simulation using the FFT solver. The red solid curve corresponded to the theoretical dispersion relation for IAW in a hot, unmagnetized plasma, under the condition $T_e \gg T_i$

$$\frac{\omega^2}{k^2} = \frac{\kappa T_e}{m_i} \frac{1}{1 + k^2 \lambda_{De}^2}, \quad (6)$$

where T_e is the electron temperature, T_i is the ion temperature, κ is the Boltzmann constant, and λ_{De} represents the Debye length.¹⁹ As shown, the modes in Fig. 4(a) aligned closely with the theoretical dispersion relation.

To assess the impact of RSC and BLS, we performed four additional simulations using the hardware solver, with variations involving RSC and BLS. The baseline for the electron and ion densities was set to the initial background distribution function when BLS was enabled.

We compared the time history of $|E_x^{3,0,0}|$ across different simulations in Fig. 4(b). When neither RSC nor BLS was applied, the results (the orange line) exhibited significant deviations from the reference double-precision result (the blue line) and noticeable noises. The resulting wave spectra did not align well with the theoretic dispersion relation [Fig. 4(c)]. Results with RSC but without BLS (the yellow line)

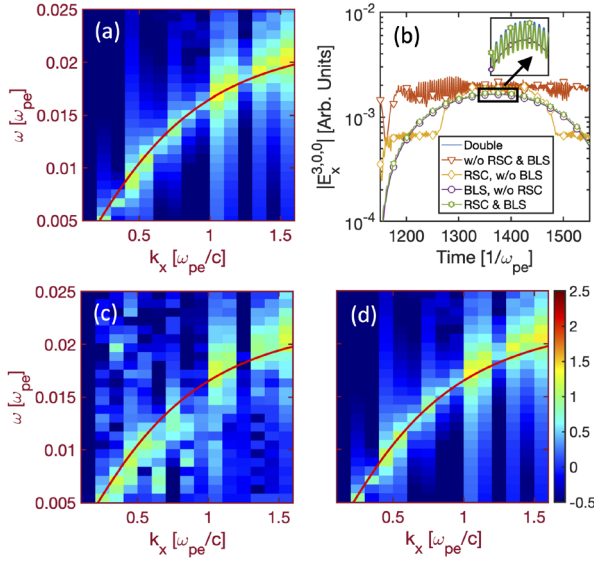


FIG. 4. Wave spectra of $\log_{10}|E_x(k_x, \omega)|$ in a hot, unmagnetized plasma: (a) Using the FFT solver, (c) using the hardware solver without optimization, and (d) using the hardware solver with RSC and BLS. The red solid curve represents the theoretical dispersion relation from Eq. (6). Subfigures (a), (c), and (d) share the same colorbar. (b) Comparison of the time history of $|E_x^{3,0,0}|$ across different simulations. In the legend, “LP” represents limited precision, “RSC” denotes rescaling, and “BLS” stands for baseline shifting.

showed improved performance but still deviated notably from the reference when $|E_x^{3,0,0}|$ was smaller than 10^{-3} . This is expected, as the amplitudes of the perturbations are important for accurately capturing the IAWs in this problem. Without BLS, the scale of the distribution function coefficients was determined by the dominant term $C_{s,0,0,0}^{0,0,0}$, which remained unchanged in the simulation.

Results with BLS but without RSC (the purple line) tracked closer to the reference curve but were less accurate at the electron plasma frequency scale, as highlighted in the zoomed-in inset. This discrepancy arose because the fluctuations at the electron plasma frequency became much weaker over time due to strong electron Landau damping. BLS alone cannot sufficiently improve the code’s performance at such low amplitudes.

In contrast, when both RSC and BLS were applied (the green line), the results closely matched the double-precision reference curve, with excellent agreement in the zoomed-in region. The corresponding wave spectra [Fig. 4(d)] were also nearly identical to the reference [Fig. 4(a)], showing the combined effectiveness of RSC and BLS in achieving accuracy comparable to the double-precision solution when using the hardware solver.

E. Two-stream instabilities in the linear stage

To study *SpectraVla*’s ability to model instabilities with growing amplitudes, we chose to model the two-stream instability (TSI) from two counter-propagating electron species with the same density. The theoretical growth rate γ of the TSI modes satisfies the following expression:²⁰

$$\gamma^2 = \sqrt{1 + \frac{8k^2 u_e^2}{\omega_{pe}^2}} - \left(1 + \frac{2k^2 u_e^2}{\omega_{pe}^2}\right). \quad (7)$$

Below we use the superscript $\pm$ to distinguish the two electron species. The parameters used for these 1D1V simulations were $L = \{32\pi, 32\pi, 32\pi\}$, $n_x = \{256, 1, 1\}$, $n_y = \{300, 1, 1\}$, $\alpha_e^+ = \alpha_e^- = \{0.1, 1.0, 1.0\}$, $u_e^+ = -u_e^- = \{1.0, 0.0, 0.0\}$, $m_i/m_e = 1836.0$, $\alpha_i = \{0.1\sqrt{m_e/m_i}, 1.0, 1.0\}$, $u_i = \{0.0, 0.0, 0.0\}$, $\omega_{pe}/\omega_{ce} = 1.0$, $C_{e,0,0,0}^+ = C_{e,0,0,0}^- = 0.5/(\alpha_{e,x}\alpha_{e,y}\alpha_{e,z})$, and $C_{i,0,0,0} = 1/(\alpha_{i,x}\alpha_{i,y}\alpha_{i,z})$. At $t = 0$, the system was perturbed with uniform random noises ranging from $-\epsilon$ to ϵ in E_x , where $\epsilon = 10^{-4}$. The simulations ran up to $\omega_{pe}t = 25.0$ with a time step of $\omega_{pe}dt = 0.1$ to obtain the linear growth rate of TSI modes. An artificial damping rate $\mu = 5.0\omega_{pe}$ was used.

Five simulations were conducted—one using the FFT solver and four using the hardware solver, with variations involving RSC and BLS. The baseline for the electron and ion densities was set to the background density when BLS was enabled.

Figure 5(a) shows the mean relative error of E_x between $0.375k_0$ and $0.813k_0$ ($k_0 = \omega_{pe}/c$) from the hardware solver compared to the FFT solver. Overall, the hardware solver with both RSC and BLS techniques performed the best, maintaining a mean relative error below 0.2%. In contrast, the hardware solver with neither RSC nor BLS exhibited the poorest performance, with the mean relative errors reaching up to 100%. When only one technique was enabled, the hardware solver still delivered good predictions: the mean relative error remained below 2% with RSC and below 0.9% with BLS. Notably, the BLS technique performed better at the beginning of the simulation compared to RSC.

Accurately modeling TSI in the linear stage required calculating the convolution term $C_{s,n,0,0} * E_x \equiv C_{s,n,0,0}^b * E_x + dC_{s,n,0,0} * E_x$. In the linear stage, the second term $dC_{s,n,0,0} * E_x$ was a second-order small quantity and can be neglected. RSC improved the calculation of $C_{s,n,0,0}^b * E_x$ by scaling E_x to values that the hardware can more accurately process. BLS treated this term as linear and computed it without the convolution solver. Both RSC and BLS were capable of independently solving the physics of the linear stage. However, BLS provided more accurate results for $dC_{s,n,0,0} * E_x$. When only RSC was enabled, the hardware was unable to capture $dC_{s,n,0,0}$, resulting in $dC_{s,n,0,0} * E_x$ being 0.

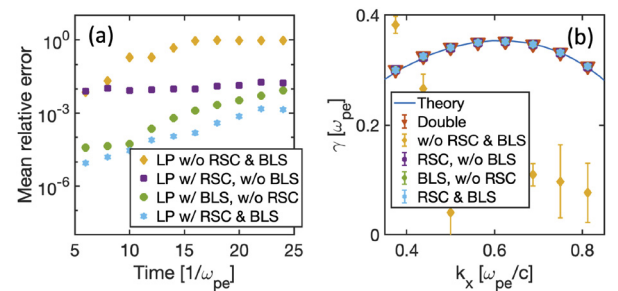


FIG. 5. The two-stream instability simulation results. (a) The mean relative error of E_x using different solvers and precision enhancements. (b) Comparison of the growth rate from the simulations and the theory. In the legend, “LP” represents limited precision, “RSC” denotes rescaling, and “BLS” stands for baseline shifting.

Figure 5(b) shows the linear growth rates obtained, along with their error bars. Four of the simulations accurately predicted the linear growth rates of the dominant TSI modes, with uncertainties within $\pm 10^{-3} \omega_{pe}$. The simulation utilizing the hardware solver without RSC or BLS produced a poor prediction.

F. Two-stream instabilities in the nonlinear stage

To further evaluate the performance of the hardware solver, we simulated a single-mode TSI into the nonlinear regime. The parameters used for these 1D1V simulations were $L = \{2\pi, 2\pi, 2\pi\}$, $n_x = \{64, 1, 1\}$, $n_y = \{500, 1, 1\}$, $\alpha_e^+ = \alpha_e^- = \{0.5, 1.0, 1.0\}$, $u_e^+ = -u_e^- = \{1.0, 0.0, 0.0\}$, $m_i/m_e = 1836.0$, $\alpha_i = \{0.5\sqrt{m_e/m_i}, 1.0, 1.0\}$, $u_i = \{0.0, 0.0, 0.0\}$, $\omega_{pe}/\omega_{ce} = 1.0$, $C_{e,0,0,0}^+ = C_{e,0,0,0}^- = 0.5/(\alpha_{e,x}\alpha_{e,y}\alpha_{e,z})$, and $C_{i,0,0,0} = 1/(\alpha_{i,x}\alpha_{i,y}\alpha_{i,z})$. At $t = 0$, the system was perturbed with $E_x = \epsilon \cos(x)$, where $\epsilon = 0.001$. The simulations ran up to $\omega_{pe}t = 100.0$ with a time step of $\omega_{pe}dt = 0.1$. An artificial damping rate $\mu = 5.0\omega_{pe}$ was used.

We performed two simulations: one using the FFT solver and the other utilizing the hardware solver with both BLS and RSC enabled. The baseline for the electron and ion densities was set to the background density. To enhance the resolution of the fine structures within the density distribution function, we refined the RSC algorithm by replacing the previously used global constant scaling factor $C_{s, \text{scale}}$, which was applied to the entire 6D array $\{C_{s,n,m,p}^{k_x,k_y,k_z}\}$, with a more elaborate approach. We introduced a variable scaling factor $C_{s,n,m,p}^{\text{scale}}$ that is applied to each 3D sub-array $\{C_{s,n,m,p}^{k_x,k_y,k_z}\}$ for specific n , m , and p values, thus permitting distinct scaling factors for different n , m , and p combinations. This rescaling strategy was implemented without modifying other aspects of the algorithm. The improved method significantly enhanced the code's accuracy, particularly for arrays $\{C_{s,n,m,p}^{k_x,k_y,k_z}\}$ that span a broad range of scales.

Figure 6(a) shows the time history of the amplitude for the dominant mode $E_x^{1,0,0}$ obtained from the two simulations, with a mean relative error of 2.54% between the two. The electron density distribution functions at $\omega_{pe}t = 70$, as depicted in Figs. 6(b) and 6(c) from the FFT solver and the hardware solver, respectively, are nearly identical. This indicates that when employing RSC and BLS, the hardware solver can yield predictions closely approximating those from double precision simulations even in the nonlinear stage.

V. TIME BREAKDOWN

Table I provides a breakdown of the simulation times for the cases using the hardware solver with both RSC and BLS. As the convolution size increased, the computation time became increasingly dominated by the convolution operation, rising from 70% in LD to 98% in TSI (Lin.). This indicates that *SpectraVla* can potentially benefit significantly from hardware capable of fast convolution execution. Except for small convolution sizes, such as the 16 grids in the LD case, RSC and BLS, in general, only cost less than 1% of the computation time, with the percentage decreasing with the problem size increase in general. With the refined RSC algorithm implemented in the nonlinear TSI case, a higher number of RSC operations were necessary, which accounted for 2.15% of the main loop's time. Nonetheless, this figure remains negligible when compared to the time spent on convolution.

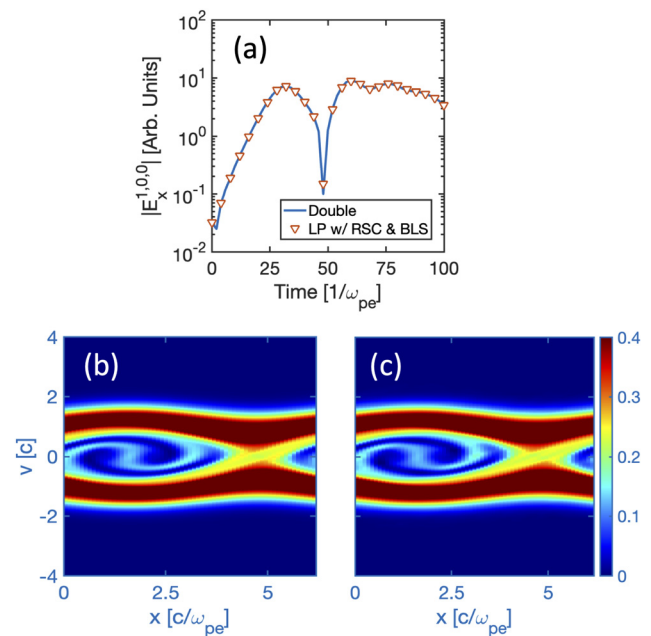


FIG. 6. The two-stream instability simulation results in the nonlinear regime: (a) The time evolution of the dominant mode $E_x^{1,0,0}$. (b) and (c) The electron density distribution functions at $\omega_{pe}t = 70$ from simulations using (b) the FFT solver and (c) the hardware solver. In the legend, “LP” represents limited precision, “RSC” denotes rescaling, and “BLS” stands for baseline shifting.

TABLE I. Time breakdown of the simulations using the hardware solver. In this table, “ML” stands for the main loop, “Conv.” represents convolution, and “k” denotes multiplication by 10^3 . The final column shows the percentage of the convolution time relative to the total main loop time.

Problem	ML (s)	Conv. (s)	RSC (s)	BLS (s)	%Conv.
LD	19.4	13.6	0.301	0.659	70.1%
WMP (//)	934	847	0.116	8.76	90.7%
WMP ($\perp$)	931	848	0.0979	8.64	91.1%
TSI (Lin.)	1.33k	1.30k	1.45	3.57	97.7%
IAW	16.9k	15.3k	2.38	169	90.5%
TSI (Nonlin.)	460	392	9.99	6.13	85.2%

Therefore, these techniques can improve simulation accuracy without posing a bottleneck to the overall hardware speedup.

While RSC and BLS can be performed at every time step, such operations necessitate an update to the preconditioning matrix, as the coefficients for the variables will change. This process is computationally expensive and time-consuming, which is why they are not applied at every step.

VI. SUMMARY AND DISCUSSION

In summary, we presented the code *SpectraVla*, which solves multi-dimensional Vlasov–Maxwell equations using the spectral method. The code has been benchmarked with theory on a number of fundamental plasma physics problems. In addition to the conventional

FFT convolution solver, a hardware convolution solver has also been developed to study the code performance on an under-development hyperspectral photonic tensor core with limited-precision numbers. By applying rescaling and baseline shifting techniques, this code can deliver results using 8-bit fixed-point numbers with accuracy comparable to double precision simulations, with minimal overhead. The photonic tensor core and related chip-scale design is under way. Actual performance of *SpectraVla* on the final hardware may depend on a number of factors, notably the data transfer speed between the tensor core and memory and further code optimization.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

S. H. Cao: Data curation (lead); Formal analysis (lead); Investigation (lead); Methodology (equal); Software (lead); Validation (lead); Visualization (lead); Writing – original draft (lead); Writing – review & editing (equal). **S. Munasinghe:** Investigation (supporting); Validation (supporting). **M. Huang:** Conceptualization (supporting); Funding acquisition (equal); Investigation (equal); Methodology (supporting); Project administration (equal); Resources (equal); Supervision (equal); Visualization (equal); Writing – review & editing (equal). **H. Wu:** Conceptualization (supporting); Funding acquisition (equal); Investigation (equal); Methodology (supporting); Project administration (equal); Resources (equal); Supervision (equal); Writing – review & editing (equal). **Q. Lin:** Conceptualization (supporting); Funding acquisition (lead); Investigation (equal); Project administration (lead); Resources (lead); Supervision (equal); Writing – review & editing (supporting). **C. Ren:** Conceptualization (lead);

Formal analysis (equal); Funding acquisition (equal); Investigation (equal); Methodology (equal); Project administration (equal); Resources (equal); Supervision (lead); Writing – review & editing (lead).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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